Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

RealMLP XAI study notebook

This notebook is a model-specific XAI study for the saved 10-fold RealMLP checkpoint family reconstructed from `churn_analysis_helpers.py`.

The focus is not merely to replay predictions. The focus is to understand the model's smooth nonlinear response geometry, feature sensitivity, and where that geometry helps or fails.

Study perspective

We treat the fitted predictor as an ensemble-averaged neural probability map:

$$p_{\mathrm{mlp}}(x) = \frac{1}{K} \sum_{k=1}^{K} softmax ¿$$

Unlike the tree model, the RealMLP combines:

- robust smooth-clipped numerical scaling,
- periodic numerical embeddings,
- one-hot and learnt categorical embeddings,
- stacked GELU nonlinearities,
- fold-model averaging.

The central XAI questions are therefore:

1. Which variables drive the ensemble's global behaviour?
2. Where is the response surface smooth, sharp, or unstable?
3. Which cases create member disagreement?
4. Which inputs move the model most under local perturbation or ablation?

Required files in `./`

- `train.csv`
- `test.csv`
- `realmlp_fold_*.pth`
- `churn_analysis_helpers.py`

```
from pathlib import Path
import json
import warnings
warnings.filterwarnings("ignore")
```

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import torch
import shap

from sklearn.metrics import roc_auc_score, log_loss, brier_score_loss
from sklearn.inspection import permutation_importance

from churn_analysis_helpers import (
    CFG,
    seed_everything,
    setup_plotting,
    ensure_output_dir,
    savefig,
    build_feature_frames,
    prepare_realmlp_frames,
    inspect_realmlp_schema,
    replay_realmlp,
    make_decile_frame,
    cumulative_capture,
    FE_realmlp,
    RobustScaleSmoothClipTransform,
    RealMLP,
    load_realmlp_checkpoints,
)
from sklearn.preprocessing import TargetEncoder

seed_everything(67)
setup_plotting()

cfg = CFG()
ensure_output_dir(cfg)

# Optional path overrides
# cfg.train_path = "./train.csv"
# cfg.test_path = "./test.csv"
# cfg.realmlp_pattern = "realmlp_fold_*.pth"
# cfg.output_dir = "./analysis_outputs_realmlp"
```

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

1. Reconstruct the exact RealMLP feature space

This section uses the helper file exactly as written to rebuild the numerical and categorical representations used by the saved checkpoints.

```
bundle = build_feature_frames(cfg)
realmlp_bundle = prepare_realmlp_frames(bundle)
schema = inspect_realmlp_schema(realmlp_bundle, cfg)

print("Numerical features:", schema["numeric_count"])
print("Categorical features:", schema["categorical_count"])
print("Target-encoded cats:", schema["target_encoded_cat_count"])
print("Augmented numeric dim:", schema["augmented_numeric_count"])
display(pd.Series(schema).to_frame("value"))
```

```
Numerical features: 22
Categorical features: 54
Target-encoded cats: 32
Augmented numeric dim: 54


value
numeric_count
22
categorical_count
54
target_encoded_cat_count
32
augmented_numeric_count
54
NUMS                        [SeniorCitizen, tenure, MonthlyCharges,
TotalC...
CATEGORY                    [gender, Partner, Dependents, PhoneService,
Mu...
```

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2. Replay the saved fold checkpoints

The helper already reconstructs the exact replay path. We use that first, then build additional XAI routines around the replayed pipeline.

```
replay = replay_realmlp(bundle, realmlp_bundle, cfg)

print("device:", replay["device"])
```

```
print("RealMLP OOF AUC:", f"{replay['realmlp_oof_auc']:.6f}")
print("RealMLP OOF log loss:", f"{log_loss(bundle['y'],
replay['realmlp_oof']):.6f}")
print("RealMLP OOF Brier:", f"{brier_score_loss(bundle['y'],
replay['realmlp_oof']):.6f}")
```

```
device: cpu
RealMLP OOF AUC: 0.917199
RealMLP OOF log loss: 0.298696
RealMLP OOF Brier: 0.095583
```

```
analysis_train = pd.DataFrame({
    cfg.id_col: bundle["train"][cfg.id_col].values,
    "y_true": bundle["y"],
    "realmlp_pred": replay["realmlp_oof"],
})

analysis_test = pd.DataFrame({
    cfg.id_col: bundle["test"][cfg.id_col].values,
    "realmlp_pred": replay["realmlp_test_mean"],
})

analysis_train["pred_rank_pct"] =
analysis_train["realmlp_pred"].rank(pct=True, method="average")
analysis_train["hard_label"] = (analysis_train["realmlp_pred"] >=
0.5).astype(int)
analysis_train["error"] = analysis_train["y_true"] -
analysis_train["realmlp_pred"]
analysis_train["abs_error"] = analysis_train["error"].abs()
analysis_train["case_type"] = np.select(
    [
        (analysis_train["y_true"] == 1) &
(analysis_train["hard_label"] == 1),
        (analysis_train["y_true"] == 0) &
(analysis_train["hard_label"] == 0),
        (analysis_train["y_true"] == 0) &
(analysis_train["hard_label"] == 1),
        (analysis_train["y_true"] == 1) &
(analysis_train["hard_label"] == 0),
    ],
    ["TP", "TN", "FP", "FN"],
    default="other"
)
analysis_train["risk_band"] = pd.qcut(
    analysis_train["realmlp_pred"], q=5,
    labels=["very_low", "low", "mid", "high", "very_high"],
    duplicates="drop",
)

member_cols = [f"member_{i}" for i in range(1,
```

```
replay["realmlp_test_members"].shape[1] + 1)]
test_member_df = pd.DataFrame(replay["realmlp_test_members"],
columns=member_cols)
analysis_test = pd.concat([analysis_test, test_member_df], axis=1)
analysis_test["member_std"] = analysis_test[member_cols].std(axis=1)
analysis_test["member_range"] = analysis_test[member_cols].max(axis=1)
- analysis_test[member_cols].min(axis=1)

analysis_train.head()
```

```
   id  y_true  realmlp_pred  pred_rank_pct  hard_label     error
abs_error  \
0   0       0      0.009655       0.282337           0 -0.009655
0.009655
1   1       0      0.002292       0.044788           0 -0.002292
0.002292
2   2       0      0.301171       0.704309           0 -0.301171
0.301171
3   3       1      0.707725       0.927854           1  0.292275
0.292275
4   4       1      0.788549       0.960841           1  0.211451
0.211451

  case_type  risk_band
0        TN        low
1        TN   very_low
2        TN       high
3        TP  very_high
4        TP  very_high
```

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3. Behavioural diagnostics

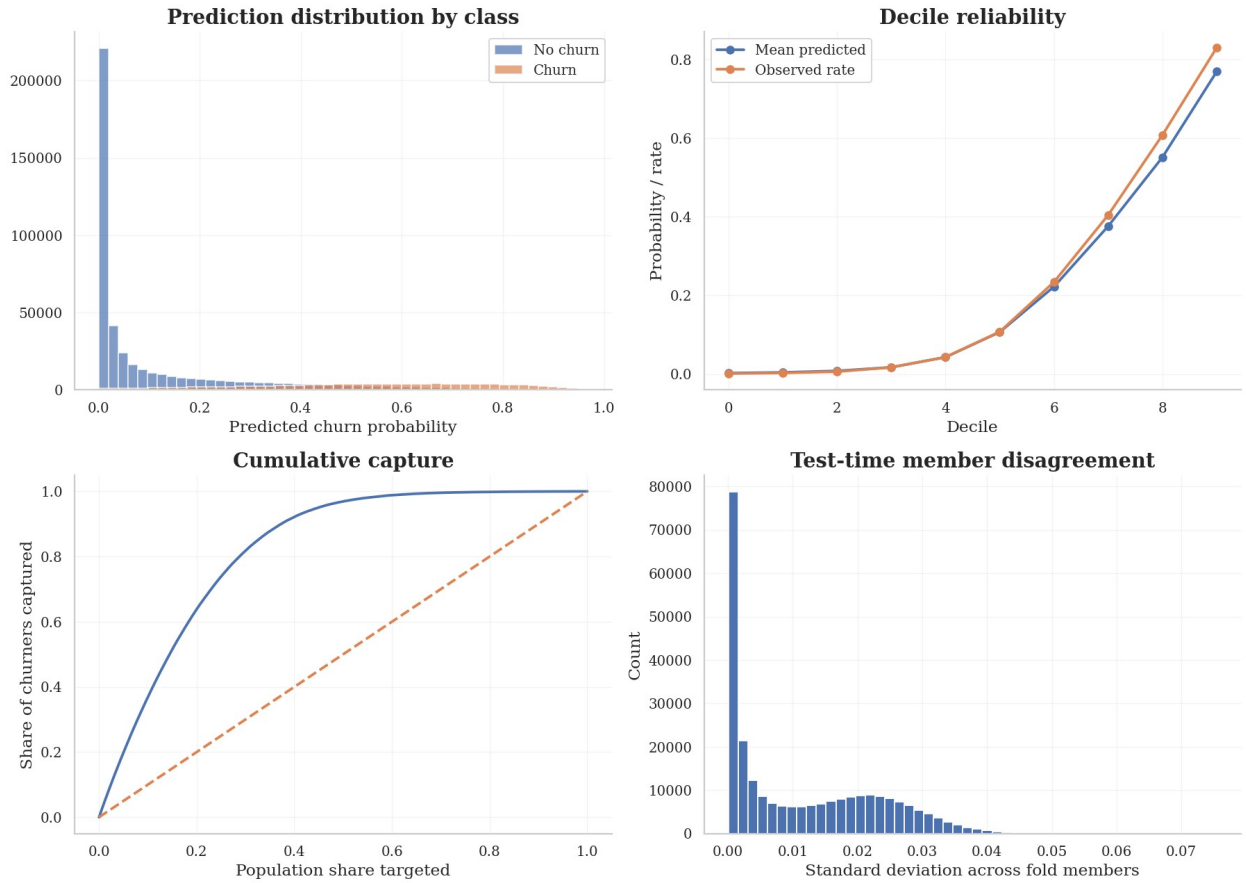
These plots answer whether the replayed ensemble:

- separates classes cleanly,
- calibrates sensibly,
- concentrates positives in the upper tail,
- produces tail overconfidence or underconfidence.

```
fig, axes = plt.subplots(2, 2, figsize=(14, 10))

axes[0, 0].hist(analysis_train.loc[analysis_train["y_true"] == 0,
"realmlp_pred"], bins=50, alpha=0.7, label="No churn")
```

```
axes[0, 0].hist(analysis_train.loc[analysis_train["y_true"] == 1,
"realmlp_pred"], bins=50, alpha=0.7, label="Churn")
axes[0, 0].set_title("Prediction distribution by class")
axes[0, 0].set_xlabel("Predicted churn probability")
axes[0, 0].legend()

dec_realmlp = make_decile_frame(bundle["y"],
analysis_train["realmlp_pred"], n_bins=10)
axes[0, 1].plot(dec_realmlp["mean_pred"], marker="o", label="Mean
predicted")
axes[0, 1].plot(dec_realmlp["churn_rate"], marker="o", label="Observed
rate")
axes[0, 1].set_title("Decile reliability")
axes[0, 1].set_xlabel("Decile")
axes[0, 1].set_ylabel("Probability / rate")
axes[0, 1].legend()

cap_realmlp = cumulative_capture(bundle["y"],
analysis_train["realmlp_pred"])
axes[1, 0].plot(cap_realmlp["population_share"],
cap_realmlp["capture_rate"])
axes[1, 0].plot([0, 1], [0, 1], linestyle="--")
axes[1, 0].set_title("Cumulative capture")
axes[1, 0].set_xlabel("Population share targeted")
axes[1, 0].set_ylabel("Share of churners captured")

axes[1, 1].hist(analysis_test["member_std"], bins=50)
axes[1, 1].set_title("Test-time member disagreement")
axes[1, 1].set_xlabel("Standard deviation across fold members")
axes[1, 1].set_ylabel("Count")

plt.tight_layout()
savefig(cfg, "realmlp_01_core_diagnostics.png")
plt.show()

display(dec_realmlp)
```

```
saved: analysis_outputs\realmlp_01_core_diagnostics.png
```



	bin	mean_pred	churn_rate	n	pred_min	pred_max
0	0	0.002330	0.001094	59420	0.000880	0.003107
1	1	0.004047	0.002710	59419	0.003107	0.005285
2	2	0.007776	0.005840	59419	0.005285	0.011012
3	3	0.017026	0.016829	59420	0.011013	0.025836
4	4	0.042658	0.042966	59419	0.025837	0.065962
5	5	0.106499	0.107390	59419	0.065962	0.156202
6	6	0.221997	0.233793	59420	0.156208	0.294355
7	7	0.375918	0.404180	59419	0.294356	0.460177
8	8	0.551160	0.607600	59419	0.460180	0.647994
9	9	0.768880	0.829670	59420	0.648000	0.966818

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

4. Build reusable prediction functions for custom slices

The helper already provides replay for the original train/test split. For XAI work we also need predictions on modified input frames.

The next cell wraps the same preprocessing logic used by the helper so that we can:

- perturb rows,
- draw two-dimensional conditional slices,
- compute finite-difference sensitivities,
- run grouped ablations.

This cell does not modify the helper file. It only uses it as-is.

```
def prepare_custom_realmlp_df(raw_df, realmlp_bundle, cfg):
    df = raw_df.copy()
    if cfg.id_col in df.columns:
        df = df.drop(columns=[cfg.id_col]).copy()

    if cfg.target_col in df.columns:
        df[cfg.target_col] = (df[cfg.target_col] == "Yes").astype(int)

    df = FE_realmlp(df, cfg)

    for c in ["MonthlyCharges", "TotalCharges"]:
        for n_bins in ([100, 1000] if c == "MonthlyCharges" else [500,
4000]):
            kb = realmlp_bundle["kb_objects"][f"{c}_bin{n_bins}"]
            vals = df[c].values.reshape(-1, 1)
            df[f"{c}_bin{n_bins}"] =
kb.transform(vals).flatten().astype(int)

    drop_cols = [c for c in realmlp_bundle["DROP"] if c in df.columns]
    df = df.drop(columns=drop_cols, errors="ignore")

    if "service_count" not in df.columns:
        from churn_analysis_helpers import REALMLP_SERVICE_COLS
        df["service_count"] = (df[REALMLP_SERVICE_COLS] ==
"Yes").sum(axis=1).astype("float32")

    category = realmlp_bundle["CATEGORY"]
    nums = realmlp_bundle["NUMS"]

    for c in category:
        if c not in df.columns:
            df[c] = "__missing__"
        mapping_info = realmlp_bundle["category_mappings"][c]
        mapping = mapping_info["mapping"]
        default = mapping_info["default"]
        df[c] = df[c].map(lambda x: mapping.get(x,
default)).astype("int32")

    for c in nums:
        if c not in df.columns:
            df[c] = 0.0
```

```
    X_num = df[nums].values.astype(np.float32)
    X_cat = df[category].values.astype(np.int64)

    return df, X_num, X_cat


def build_realmlp_fold_cache(realmlp_bundle, cfg, embed_dim=10,
n_ens=8):
    device = torch.device("cuda" if torch.cuda.is_available() else
"cpu")
    X_num = realmlp_bundle["X_num"]
    X_cat = realmlp_bundle["X_cat"]
    y_nn = realmlp_bundle["y_nn"]
    cat_dims_global = realmlp_bundle["cat_dims_global"]
    cols_to_encode = realmlp_bundle["cols_to_encode"]
    ckpts = load_realmlp_checkpoints(cfg)

    from sklearn.model_selection import StratifiedKFold
    skf = StratifiedKFold(n_splits=cfg.n_splits, shuffle=True,
random_state=cfg.seed)

    cache = []
    for fold, ((train_idx, valid_idx), ckpt_path) in
enumerate(zip(skf.split(X_num, y_nn), ckpts), start=1):
        X_num_train, X_cat_train, y_train = X_num[train_idx],
X_cat[train_idx], y_nn[train_idx]

        te = TargetEncoder(target_type="binary", cv=5)
        if len(cols_to_encode) > 0:
            X_cat_train_encoded = te.fit_transform(X_cat_train[:,
cols_to_encode], y_train)
        else:
            X_cat_train_encoded = np.zeros((len(train_idx), 0),
dtype=np.float32)

        X_num_train_aug = np.concatenate([X_num_train,
X_cat_train_encoded], axis=1)
        rssc = RobustScaleSmoothClipTransform().fit(X_num_train_aug)

        state = torch.load(ckpt_path, map_location=device)
        expected_n_num = int(state["num_embed.w1"].shape[1])

        model = RealMLP(
            input_dim=int(expected_n_num * 5 + sum([c if c <= 8 else
embed_dim for c in cat_dims_global])),
            output_dim=2,
            cat_dims=cat_dims_global,
            n_numerical=expected_n_num,
            n_ens=n_ens,
```

```
            embed_dim=embed_dim,
            device=device,
        ).to(device)
        model.load_state_dict(state)
        model.eval()

        cache.append({
            "fold": fold,
            "train_idx": train_idx,
            "valid_idx": valid_idx,
            "te": te,
            "rssc": rssc,
            "model": model,
            "device": device,
        })
    return cache


def predict_realmlp_custom(raw_df, realmlp_bundle, cfg, fold_cache,
eval_bs=8192):
    _, X_num_custom, X_cat_custom = prepare_custom_realmlp_df(raw_df,
realmlp_bundle, cfg)
    cols_to_encode = realmlp_bundle["cols_to_encode"]

    preds = []
    member_preds = []
    for item in fold_cache:
        te = item["te"]
        rssc = item["rssc"]
        model = item["model"]
        device = item["device"]

        if len(cols_to_encode) > 0:
            X_cat_encoded = te.transform(X_cat_custom[:,
cols_to_encode])
        else:
            X_cat_encoded = np.zeros((len(X_num_custom), 0),
dtype=np.float32)

        X_num_aug = np.concatenate([X_num_custom, X_cat_encoded],
axis=1)
        X_num_aug = rssc.transform(X_num_aug)

        X_num_t = torch.as_tensor(X_num_aug, dtype=torch.float32,
device=device)
        X_cat_t = torch.as_tensor(X_cat_custom, dtype=torch.long,
device=device)

        fold_probs = []
        with torch.no_grad():
```

```
            for start in range(0, len(raw_df), eval_bs):
                probs = model(X_num_t[start:start + eval_bs],
X_cat_t[start:start + eval_bs]).mean(dim=1)[:, 1]
                fold_probs.append(probs.detach().cpu().numpy())
        fold_probs = np.concatenate(fold_probs, axis=0)
        preds.append(fold_probs)

        # Member-level predictions from the ensemble dimension
        with torch.no_grad():
            member_block = []
            for start in range(0, len(raw_df), eval_bs):
                prob_full = model(X_num_t[start:start + eval_bs],
X_cat_t[start:start + eval_bs])[:, :, 1]
                member_block.append(prob_full.detach().cpu().numpy())
        member_block = np.concatenate(member_block, axis=0)
        member_preds.append(member_block)

    preds = np.vstack(preds)
    mean_pred = preds.mean(axis=0)

    member_preds = np.stack(member_preds, axis=0)   # fold x row x
member
    flat_member_preds = member_preds.transpose(1, 0,
2).reshape(len(raw_df), -1)

    return {
        "mean_pred": mean_pred,
        "fold_pred_matrix": preds,
        "member_pred_matrix": flat_member_preds,
        "custom_X_num": X_num_custom,
        "custom_X_cat": X_cat_custom,
    }


fold_cache = build_realmlp_fold_cache(realmlp_bundle, cfg)
print("Cached folds:", len(fold_cache))
```

```
Cached folds: 10
```

```
def normalise_realmlp_fold_cache(fold_cache):
    normalised = []

    for i, fold_obj in enumerate(fold_cache):
        keys = set(fold_obj.keys())

        model = None
        target_encoder = None
        rssc = None

        for k in ["model", "net", "estimator"]:
            if k in fold_obj:
```

```
                model = fold_obj[k]
                break

        for k in ["target_encoder", "te", "target_enc", "encoder"]:
            if k in fold_obj:
                target_encoder = fold_obj[k]
                break

        for k in ["rssc", "scaler", "transform", "num_transform"]:
            if k in fold_obj:
                rssc = fold_obj[k]
                break

        if model is None:
            raise KeyError(
                f"Fold cache entry {i} is missing a model-like key.
Found keys: {sorted(keys)}"
            )
        if target_encoder is None:
            raise KeyError(
                f"Fold cache entry {i} is missing a target-encoder-
like key. Found keys: {sorted(keys)}"
            )
        if rssc is None:
            raise KeyError(
                f"Fold cache entry {i} is missing an rssc/scaler-like
key. Found keys: {sorted(keys)}"
            )

        normalised.append({
            "model": model,
            "target_encoder": target_encoder,
            "rssc": rssc,
        })

    return normalised


fold_cache = normalise_realmlp_fold_cache(fold_cache)
print(fold_cache[0].keys())
```

```
dict_keys(['model', 'target_encoder', 'rssc'])
```

Mathematical framing

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5. Permutation importance on the replayed competition train set

For the neural model, permutation importance is a robust first-line global explanation because it does not require a tree-specific decomposition and respects the actual fitted predictor.

```
from sklearn.metrics import roc_auc_score

def _fold_get(fold_obj, *names):
    for name in names:
        if name in fold_obj:
            return fold_obj[name]
    raise KeyError(f"None of {names} found in fold cache keys:
{sorted(fold_obj.keys())}")


def _build_realmlp_cached_fold_inputs(
    realmlp_bundle,
    fold_cache,
    row_idx=None,
):
    X_num = realmlp_bundle["X_num"].astype(np.float32)
    X_cat = realmlp_bundle["X_cat"].astype(np.int64)
    y = realmlp_bundle["y_nn"]
    cols_to_encode = realmlp_bundle["cols_to_encode"]

    if row_idx is None:
        row_idx = np.arange(X_num.shape[0])

    X_num = X_num[row_idx].copy()
    X_cat = X_cat[row_idx].copy()
    y = y[row_idx].copy()

    cached = []
    for fold_obj in fold_cache:
        te = _fold_get(fold_obj, "target_encoder", "te", "target_enc",
"encoder")
        rssc = _fold_get(fold_obj, "rssc", "scaler", "transform",
"num_transform")
        model = _fold_get(fold_obj, "model", "net", "estimator")

        if len(cols_to_encode) > 0:
            X_cat_encoded = te.transform(X_cat[:,
cols_to_encode]).astype(np.float32)
        else:
            X_cat_encoded = np.zeros((X_num.shape[0], 0),
dtype=np.float32)

        X_num_aug = np.concatenate([X_num, X_cat_encoded],
```

```
axis=1).astype(np.float32)
        X_num_aug = rssc.transform(X_num_aug).astype(np.float32)

        cached.append({
            "model": model,
            "X_num_aug": X_num_aug,
            "X_cat": X_cat,
        })

    return cached, y, row_idx


def _predict_realmlp_from_cached(
    cached_fold_inputs,
    eval_bs=32768,
):
    all_fold_preds = []

    for fold_obj in cached_fold_inputs:
        model = fold_obj["model"]
        X_num_aug = fold_obj["X_num_aug"]
        X_cat = fold_obj["X_cat"]

        device = next(model.parameters()).device

        X_num_t = torch.as_tensor(X_num_aug, dtype=torch.float32,
device=device)
        X_cat_t = torch.as_tensor(X_cat, dtype=torch.long,
device=device)

        preds = []
        model.eval()
        with torch.no_grad():
            for start in range(0, X_num_t.shape[0], eval_bs):
                batch_num = X_num_t[start:start + eval_bs]
                batch_cat = X_cat_t[start:start + eval_bs]
                probs = model(batch_num, batch_cat).mean(dim=1)[:, 1]
                preds.append(probs.detach().cpu().numpy())

        all_fold_preds.append(np.concatenate(preds, axis=0))

    all_fold_preds = np.column_stack(all_fold_preds)
    mean_pred = all_fold_preds.mean(axis=1)
    std_pred = all_fold_preds.std(axis=1)

    return {
        "member_preds": all_fold_preds,
        "mean_pred": mean_pred,
        "std_pred": std_pred,
    }
```

```
def realmlp_numeric_permutation_importance_fast(
    realmlp_bundle,
    fold_cache,
    n_repeats=2,
    sample_n=120_000,
    random_state=67,
    eval_bs=32768,
):
    X_num_full = realmlp_bundle["X_num"]
    feature_names = list(realmlp_bundle["NUMS"])
    rng = np.random.RandomState(random_state)

    if sample_n is not None and sample_n < X_num_full.shape[0]:
        row_idx = rng.choice(X_num_full.shape[0], size=sample_n,
replace=False)
        row_idx = np.sort(row_idx)
    else:
        row_idx = np.arange(X_num_full.shape[0])

    cached_fold_inputs, y, row_idx =
_build_realmlp_cached_fold_inputs(
        realmlp_bundle=realmlp_bundle,
        fold_cache=fold_cache,
        row_idx=row_idx,
    )

    baseline = _predict_realmlp_from_cached(
        cached_fold_inputs,
        eval_bs=eval_bs,
    )["mean_pred"]
    baseline_auc = roc_auc_score(y, baseline)

    # number of original numerical features before target-encoded
columns are appended
    n_model_num_features = realmlp_bundle["X_num"].shape[1]

    rows = []
    for j, feat in enumerate(feature_names):
        aucs = []

        for _ in range(n_repeats):
            perm_idx = rng.permutation(len(y))

            permuted_cached = []
            for fold_obj in cached_fold_inputs:
                X_num_aug_perm = fold_obj["X_num_aug"].copy()
                # permute only the original numerical column j; keep
encoded cols fixed
```

```
                X_num_aug_perm[:, j] = X_num_aug_perm[perm_idx, j]

                permuted_cached.append({
                    "model": fold_obj["model"],
                    "X_num_aug": X_num_aug_perm,
                    "X_cat": fold_obj["X_cat"],
                })

            pred_perm = _predict_realmlp_from_cached(
                permuted_cached,
                eval_bs=eval_bs,
            )["mean_pred"]

            aucs.append(roc_auc_score(y, pred_perm))

        aucs = np.asarray(aucs, dtype=float)
        rows.append({
            "feature": feat,
            "baseline_auc": baseline_auc,
            "perm_auc_mean": aucs.mean(),
            "perm_auc_std": aucs.std(ddof=0),
            "mean_importance": baseline_auc - aucs.mean(),
            "std_importance": aucs.std(ddof=0),
        })

        print(
            f"{j+1:>3d}/{len(feature_names)}  {feat:<30s}  "
            f"AUC drop = {baseline_auc - aucs.mean():.6f}"
        )

    out = (
        pd.DataFrame(rows)
        .sort_values("mean_importance", ascending=False)
        .reset_index(drop=True)
    )
    return out, baseline_auc, row_idx


perm_imp, baseline_auc, row_idx =
realmlp_numeric_permutation_importance_fast(
    realmlp_bundle=realmlp_bundle,
    fold_cache=fold_cache,
    n_repeats=2,          # was 5
    sample_n=120_000,     # set None for full data
    random_state=cfg.seed,
    eval_bs=32768,        # larger batches = faster if memory allows
)

print(f"Baseline sampled OOF AUC from cached replay path:
{baseline_auc:.6f}")
```

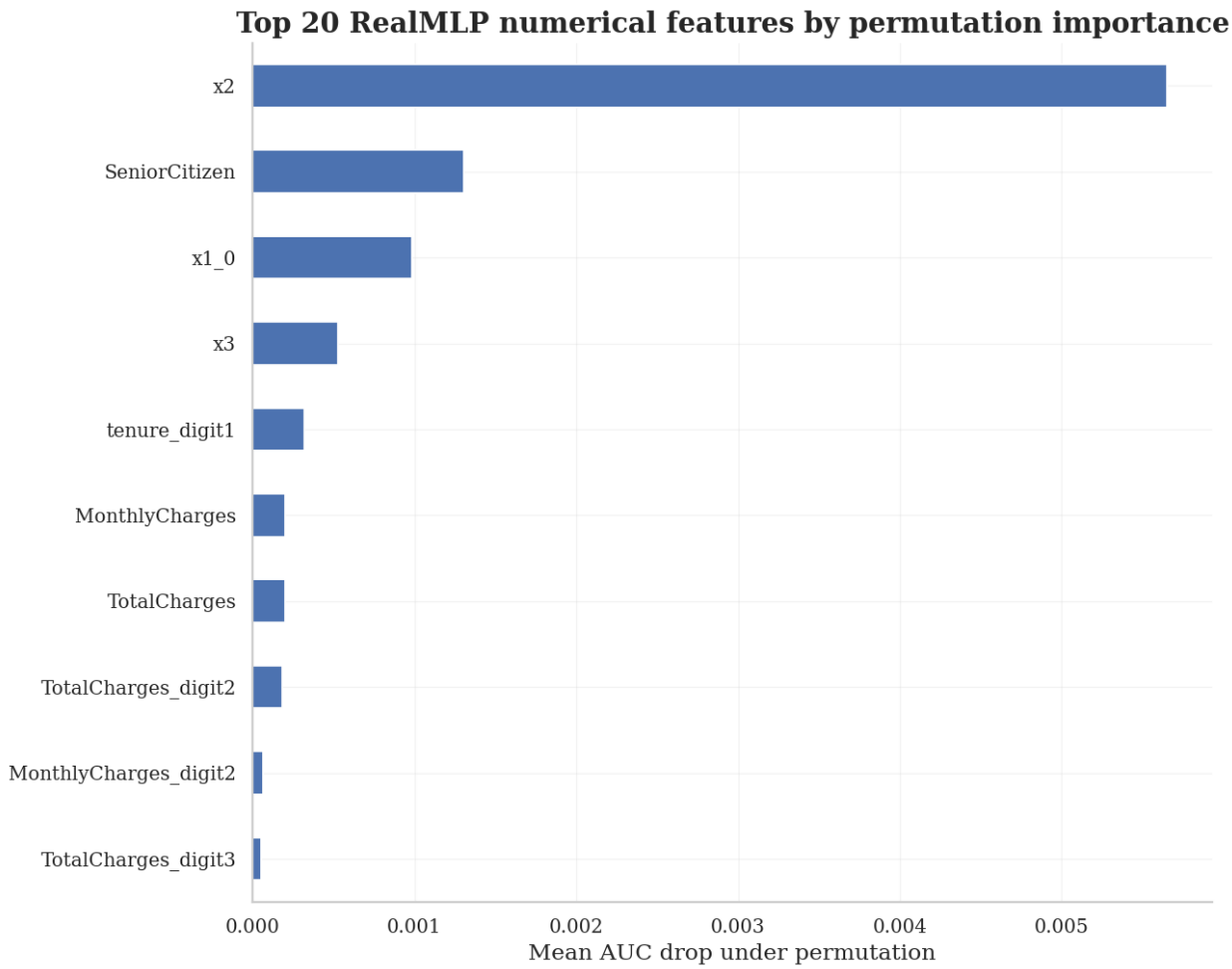
```python
print(f"Rows used: {len(row_idx):,}")
display(perm_imp.head(25))

ax = perm_imp.head(10).sort_values("mean_importance").plot(
    x="feature",
    y="mean_importance",
    kind="barh",
    legend=False,
    figsize=(10, 8),
)
ax.set_title("Top 20 RealMLP numerical features by permutation
importance")
ax.set_xlabel("Mean AUC drop under permutation")
ax.set_ylabel("")
savefig(cfg, "realmlp_02_permutation_importance.png")
plt.show()
```

```
  1/22  SeniorCitizen                    AUC drop = 0.001306
  2/22  tenure                           AUC drop = 0.000053
  3/22  MonthlyCharges                   AUC drop = 0.000202
  4/22  TotalCharges                     AUC drop = 0.000198
  5/22  tenure_digit0                    AUC drop = 0.000022
  6/22  tenure_digit1                    AUC drop = 0.000319
  7/22  MonthlyCharges_digit-2           AUC drop = 0.000012
  8/22  MonthlyCharges_digit-1           AUC drop = 0.000013
  9/22  MonthlyCharges_digit0            AUC drop = 0.000009
 10/22  MonthlyCharges_digit1            AUC drop = 0.000029
 11/22  MonthlyCharges_digit2            AUC drop = 0.000066
 12/22  TotalCharges_digit-2             AUC drop = 0.000010
 13/22  TotalCharges_digit-1             AUC drop = 0.000013
 14/22  TotalCharges_digit0              AUC drop = 0.000014
 15/22  TotalCharges_digit1              AUC drop = 0.000045
 16/22  TotalCharges_digit2              AUC drop = 0.000185
 17/22  TotalCharges_digit3              AUC drop = 0.000055
 18/22  x1_0                             AUC drop = 0.000986
 19/22  x1_1                             AUC drop = 0.000051
 20/22  x2                               AUC drop = 0.005648
 21/22  x3                               AUC drop = 0.000530
 22/22  x4                               AUC drop = 0.000030
Baseline sampled OOF AUC from cached replay path: 0.918749
Rows used: 120,000

                   feature  baseline_auc  perm_auc_mean  perm_auc_std
\
0                       x2      0.918749       0.913101  1.381999e-04

1            SeniorCitizen      0.918749       0.917442  3.250889e-05

2                     x1_0      0.918749       0.917763  9.978155e-06
```

3	x3	0.918749	0.918219	9.073667e-06
4	tenure_digit1	0.918749	0.918430	4.806324e-05
5	MonthlyCharges	0.918749	0.918547	1.387151e-05
6	TotalCharges	0.918749	0.918551	1.026089e-05
7	TotalCharges_digit2	0.918749	0.918564	1.885185e-05
8	MonthlyCharges_digit2	0.918749	0.918683	1.561111e-05
9	TotalCharges_digit3	0.918749	0.918694	9.975575e-06
10	tenure	0.918749	0.918696	8.026072e-06
11	x1_1	0.918749	0.918698	5.403114e-06
12	TotalCharges_digit1	0.918749	0.918704	1.795282e-07
13	x4	0.918749	0.918719	6.388782e-06
14	MonthlyCharges_digit1	0.918749	0.918719	4.799625e-06
15	tenure_digit0	0.918749	0.918727	1.408537e-06
16	TotalCharges_digit0	0.918749	0.918735	1.842819e-06
17	MonthlyCharges_digit-1	0.918749	0.918735	3.261082e-07
18	TotalCharges_digit-1	0.918749	0.918736	2.908277e-06
19	MonthlyCharges_digit-2	0.918749	0.918737	1.989796e-06
20	TotalCharges_digit-2	0.918749	0.918739	2.543564e-07
21	MonthlyCharges_digit0	0.918749	0.918739	1.537353e-06

	mean_importance	std_importance
0	0.005648	1.381999e-04
1	0.001306	3.250889e-05
2	0.000986	9.978155e-06
3	0.000530	9.073667e-06
4	0.000319	4.806324e-05
5	0.000202	1.387151e-05
6	0.000198	1.026089e-05
7	0.000185	1.885185e-05
8	0.000066	1.561111e-05
9	0.000055	9.975575e-06
10	0.000053	8.026072e-06

```
11          0.000051    5.403114e-06
12          0.000045    1.795282e-07
13          0.000030    6.388782e-06
14          0.000029    4.799625e-06
15          0.000022    1.408537e-06
16          0.000014    1.842819e-06
17          0.000013    3.261082e-07
18          0.000013    2.908277e-06
19          0.000012    1.989796e-06
20          0.000010    2.543564e-07
21          0.000009    1.537353e-06

saved: analysis_outputs\realmlp_02_permutation_importance.png
```




```
raw_train_no_id =
realmlp_bundle["train_nn"].drop(columns=[cfg.id_col],
errors="ignore").copy()
```

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

6. Local finite-difference sensitivity

We approximate local sensitivity of the probability surface with finite differences:

$$S_j(x) \approx \frac{p(x+he_j) - p(x-he_j)}{2h}.$$

This is not a gradient through the full preprocessing graph. It is a practical response derivative on the reconstructed end-to-end predictor.

```
def finite_difference_sensitivity_arrays(
    realmlp_bundle,
    fold_cache,
    features,
    n_rows=1500,
    row_idx=None,
    eval_bs=32768,
):
    X_num_full = realmlp_bundle["X_num"].astype(np.float32)
    X_cat_full = realmlp_bundle["X_cat"].astype(np.int64)
    feature_names = list(realmlp_bundle["NUMS"])

    feat_to_idx = {f: i for i, f in enumerate(feature_names)}
    valid_features = [f for f in features if f in feat_to_idx]

    if row_idx is None:
        row_idx = np.arange(min(n_rows, X_num_full.shape[0]))
    else:
        row_idx = np.asarray(row_idx)[:n_rows]

    X_num = X_num_full[row_idx].copy()
    X_cat = X_cat_full[row_idx].copy()

    base_out = predict_realmlp_from_arrays(
        X_num, X_cat, realmlp_bundle, fold_cache, eval_bs=eval_bs
    )
    base_pred = base_out["mean_pred"]

    rows = []
    for feat in valid_features:
        j = feat_to_idx[feat]
        col = X_num[:, j].astype(np.float64)

        sd = float(np.std(col, ddof=0))
```

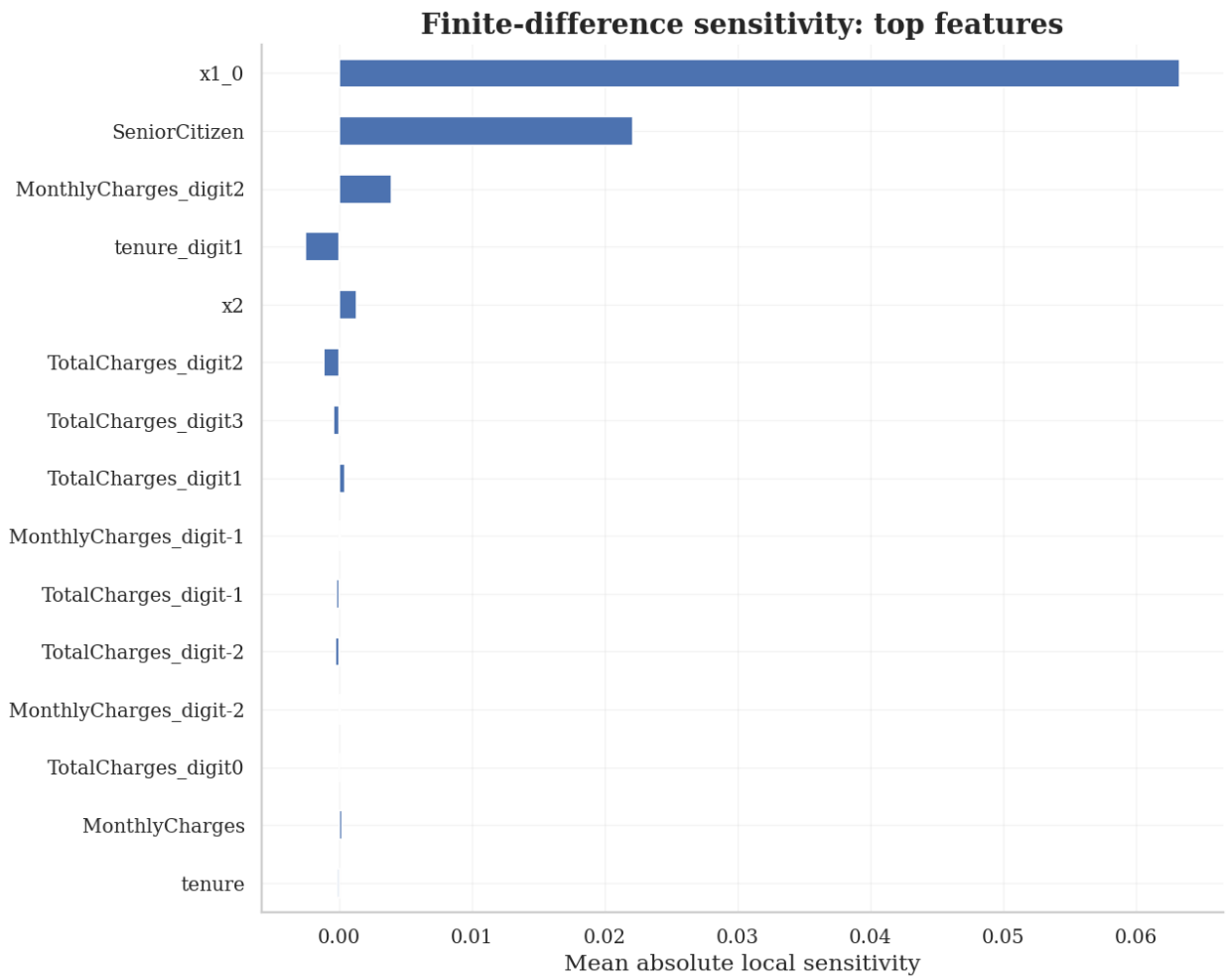
```
        step = max(sd * 0.05, 1e-3)

        X_num_plus = X_num.copy()
        X_num_minus = X_num.copy()
        X_num_plus[:, j] = X_num_plus[:, j] + step
        X_num_minus[:, j] = X_num_minus[:, j] - step

        plus_pred = predict_realmlp_from_arrays(
            X_num_plus, X_cat, realmlp_bundle, fold_cache,
eval_bs=eval_bs
        )["mean_pred"]
        minus_pred = predict_realmlp_from_arrays(
            X_num_minus, X_cat, realmlp_bundle, fold_cache,
eval_bs=eval_bs
        )["mean_pred"]

        sens = (plus_pred - minus_pred) / (2.0 * step)

        rows.append({
            "feature": feat,
            "mean_abs_sensitivity": float(np.mean(np.abs(sens))),
            "mean_signed_sensitivity": float(np.mean(sens)),
            "step": float(step),
            "base_pred_mean": float(np.mean(base_pred)),
        })

        print(
            f"{len(rows):>2d}/{len(valid_features)}  {feat:<30s}  "
            f"|mean sens| = {np.mean(np.abs(sens)):.6f}"
        )

    return pd.DataFrame(rows).sort_values("mean_abs_sensitivity",
ascending=False).reset_index(drop=True)


numeric_candidates = list(realmlp_bundle["NUMS"])[:20]

sens_df = finite_difference_sensitivity_arrays(
    realmlp_bundle=realmlp_bundle,
    fold_cache=fold_cache,
    features=numeric_candidates,
    n_rows=1500,
    eval_bs=32768,
)

display(sens_df.head(20))

ax = sens_df.head(15).sort_values("mean_abs_sensitivity").plot(
    x="feature",
    y="mean_signed_sensitivity",
```

```
    kind="barh",
    legend=False,
    figsize=(10, 8),
)
ax.set_title("Finite-difference sensitivity: top features")
ax.set_xlabel("Mean absolute local sensitivity")
ax.set_ylabel("")
savefig(cfg, "realmlp_03_finite_difference_sensitivity.png")
plt.show()
```

```
 1/20  SeniorCitizen                    |mean sens| = 0.022135
 2/20  tenure                           |mean sens| = 0.000223
 3/20  MonthlyCharges                   |mean sens| = 0.000229
 4/20  TotalCharges                     |mean sens| = 0.000003
 5/20  tenure_digit0                    |mean sens| = 0.000140
 6/20  tenure_digit1                    |mean sens| = 0.002566
 7/20  MonthlyCharges_digit-2           |mean sens| = 0.000292
 8/20  MonthlyCharges_digit-1           |mean sens| = 0.000427
 9/20  MonthlyCharges_digit0            |mean sens| = 0.000200
10/20  MonthlyCharges_digit1            |mean sens| = 0.000220
11/20  MonthlyCharges_digit2            |mean sens| = 0.003959
12/20  TotalCharges_digit-2             |mean sens| = 0.000306
13/20  TotalCharges_digit-1             |mean sens| = 0.000342
14/20  TotalCharges_digit0              |mean sens| = 0.000290
15/20  TotalCharges_digit1              |mean sens| = 0.000673
16/20  TotalCharges_digit2              |mean sens| = 0.001161
17/20  TotalCharges_digit3              |mean sens| = 0.000694
18/20  x1_0                             |mean sens| = 0.109892
19/20  x1_1                             |mean sens| = 0.000152
20/20  x2                               |mean sens| = 0.001274
```

```
                    feature  mean_abs_sensitivity
mean_signed_sensitivity  \
0                      x1_0              0.109892
0.063291
1            SeniorCitizen              0.022135
0.022135
2    MonthlyCharges_digit2              0.003959
0.003959
3            tenure_digit1              0.002566                 -
0.002566
4                       x2              0.001274
0.001268
5      TotalCharges_digit2              0.001161                 -
0.001143
6      TotalCharges_digit3              0.000694                 -
0.000433
7      TotalCharges_digit1              0.000673
0.000463
8   MonthlyCharges_digit-1              0.000427
```

```
0.000064
9     TotalCharges_digit-1              0.000342                -
0.000215
10    TotalCharges_digit-2              0.000306                -
0.000301
11  MonthlyCharges_digit-2              0.000292
0.000061
12     TotalCharges_digit0              0.000290
0.000002
13          MonthlyCharges              0.000229
0.000221
14                  tenure              0.000223                -
0.000173
15   MonthlyCharges_digit1              0.000220                -
0.000177
16   MonthlyCharges_digit0              0.000200
0.000063
17                    x1_1              0.000152                -
0.000108
18           tenure_digit0              0.000140                -
0.000051
19            TotalCharges              0.000003                -
0.000001

          step  base_pred_mean
0     0.016422        0.245557
1     0.015603        0.245557
2     0.017033        0.245557
3     0.118060        0.245557
4     1.907246        0.245557
5     0.144354        0.245557
6     0.113957        0.245557
7     0.138634        0.245557
8     0.140445        0.245557
9     0.143612        0.245557
10    0.054480        0.245557
11    0.053940        0.245557
12    0.151424        0.245557
13    1.545950        0.245557
14    1.288208        0.245557
15    0.122585        0.245557
16    0.103847        0.245557
17    1.290822        0.245557
18    0.151658        0.245557
19  116.076404        0.245557

saved: analysis_outputs\realmlp_03_finite_difference_sensitivity.png
```



Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

7. Grouped categorical ablation

For the categorical branch we use grouped ablations. A group is neutralised by setting its columns to a reference category profile and measuring the induced probability shift.

This directly answers which categorical families the neural model is leaning on most.

```
def categorical_group_ablation_arrays(
    realmlp_bundle,
    fold_cache,
    cat_groups,
    n_rows=3000,
    eval_bs=32768,
```

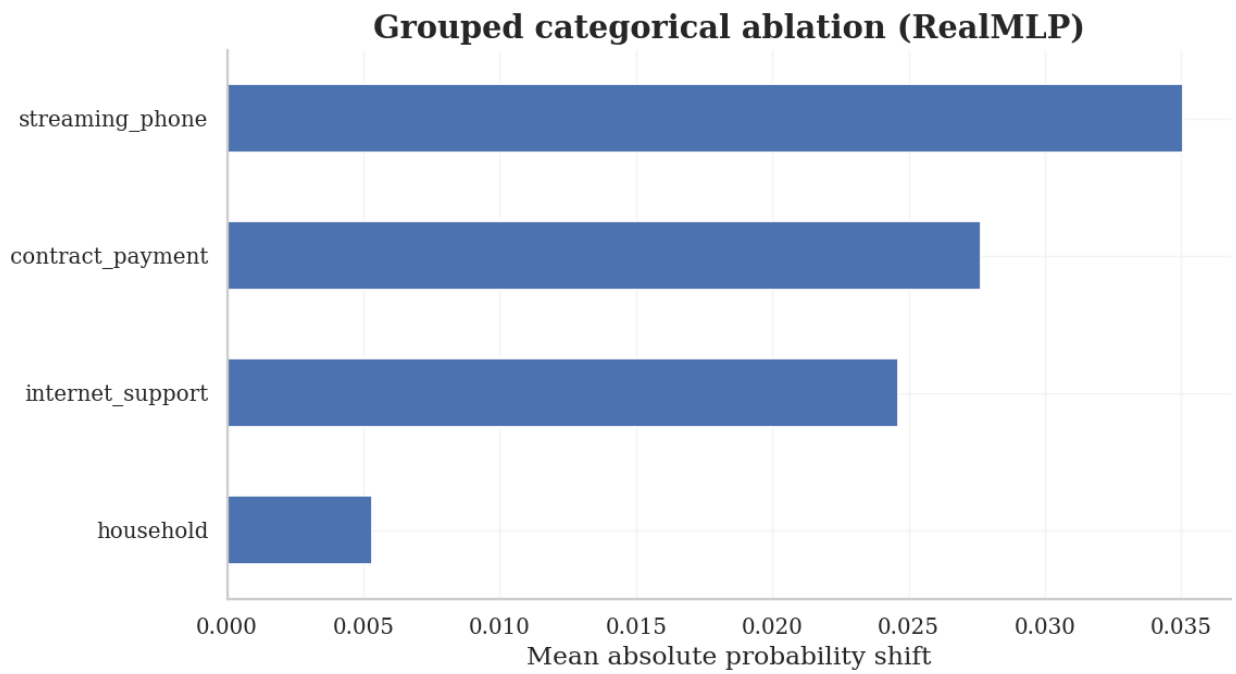
```
):
    X_num_full = realmlp_bundle["X_num"].astype(np.float32)
    X_cat_full = realmlp_bundle["X_cat"].astype(np.int64)
    y = realmlp_bundle["y_nn"]

    CATEGORY = realmlp_bundle["CATEGORY"]

    cat_to_idx = {c: i for i, c in enumerate(CATEGORY)}

    row_idx = np.arange(min(n_rows, X_num_full.shape[0]))
    X_num = X_num_full[row_idx].copy()
    X_cat = X_cat_full[row_idx].copy()

    base_pred = predict_realmlp_from_arrays(
        X_num, X_cat, realmlp_bundle, fold_cache, eval_bs=eval_bs
    )["mean_pred"]

    ablation_rows = []

    for group_name, cols in cat_groups.items():
        idxs = [cat_to_idx[c] for c in cols if c in cat_to_idx]
        if not idxs:
            continue

        X_cat_abl = X_cat.copy()

        # replace with most frequent category (mode in encoded space)
        for j in idxs:
            values, counts = np.unique(X_cat[:, j],
return_counts=True)
            mode_val = values[np.argmax(counts)]
            X_cat_abl[:, j] = mode_val

        abl_pred = predict_realmlp_from_arrays(
            X_num, X_cat_abl, realmlp_bundle, fold_cache,
eval_bs=eval_bs
        )["mean_pred"]

        delta = base_pred - abl_pred

        ablation_rows.append({
            "group": group_name,
            "mean_abs_delta": float(np.mean(np.abs(delta))),
            "mean_signed_delta": float(np.mean(delta)),
            "max_abs_delta": float(np.max(np.abs(delta))),
        })

        print(
            f"{group_name:<20s}  |Δ| = {np.mean(np.abs(delta)):.6f}"
        )
```

```
    return pd.DataFrame(ablation_rows).sort_values("mean_abs_delta",
ascending=False)


cat_groups = {
    "contract_payment": ["Contract", "PaymentMethod",
"PaperlessBilling"],
    "internet_support": ["InternetService", "OnlineSecurity",
"OnlineBackup", "DeviceProtection", "TechSupport"],
    "streaming_phone": ["PhoneService", "MultipleLines",
"StreamingTV", "StreamingMovies"],
    "household": ["gender", "SeniorCitizen", "Partner", "Dependents"],
}

ablation_df = categorical_group_ablation_arrays(
    realmlp_bundle=realmlp_bundle,
    fold_cache=fold_cache,
    cat_groups=cat_groups,
    n_rows=3000,
    eval_bs=32768,
)

display(ablation_df)

ax = ablation_df.sort_values("mean_abs_delta").plot(
    x="group",
    y="mean_abs_delta",
    kind="barh",
    legend=False,
    figsize=(9, 5),
)
ax.set_title("Grouped categorical ablation (RealMLP)")
ax.set_xlabel("Mean absolute probability shift")
ax.set_ylabel("")
savefig(cfg, "realmlp_04_grouped_ablation.png")
plt.show()
```

```
contract_payment      |Δ| = 0.027666
internet_support      |Δ| = 0.024606
streaming_phone       |Δ| = 0.035071
household             |Δ| = 0.005301
```

	group	mean_abs_delta	mean_signed_delta	max_abs_delta
2	streaming_phone	0.035071	0.034967	0.223998
0	contract_payment	0.027666	-0.027666	0.198302
1	internet_support	0.024606	-0.024606	0.317824
3	household	0.005301	-0.004702	0.110057

```
saved: analysis_outputs\realmlp_04_grouped_ablation.png
```



Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

8. Conditional decision surfaces

For selected numerical feature pairs, we visualise two-dimensional conditional slices:

$$(x_i, x_j) \mapsto p_{\text{mlp}}(x_i, x_j, \tilde{x}_{-ij}).$$

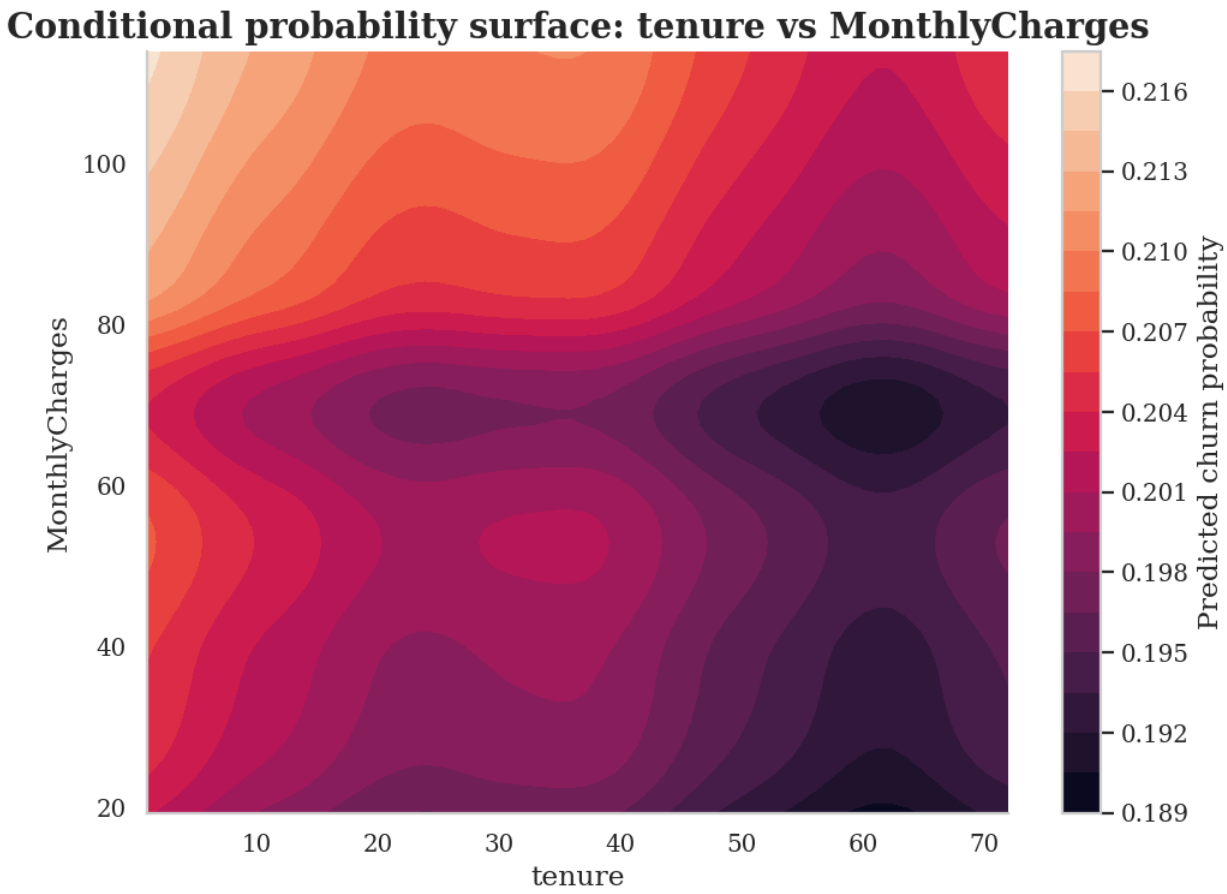
Compared with the tree model, these surfaces should be smoother and more curved if the neural representation is interpolating rather than thresholding.

```
def plot_realmlp_surface_arrays(
    realmlp_bundle,
    fold_cache,
    feature_x,
    feature_y,
    grid_n=60,
    q_low=0.02,
    q_high=0.98,
    eval_bs=32768,
):
    feature_names = list(realmlp_bundle["NUMS"])
    feat_to_idx = {f: i for i, f in enumerate(feature_names)}

    if feature_x not in feat_to_idx or feature_y not in feat_to_idx:
```

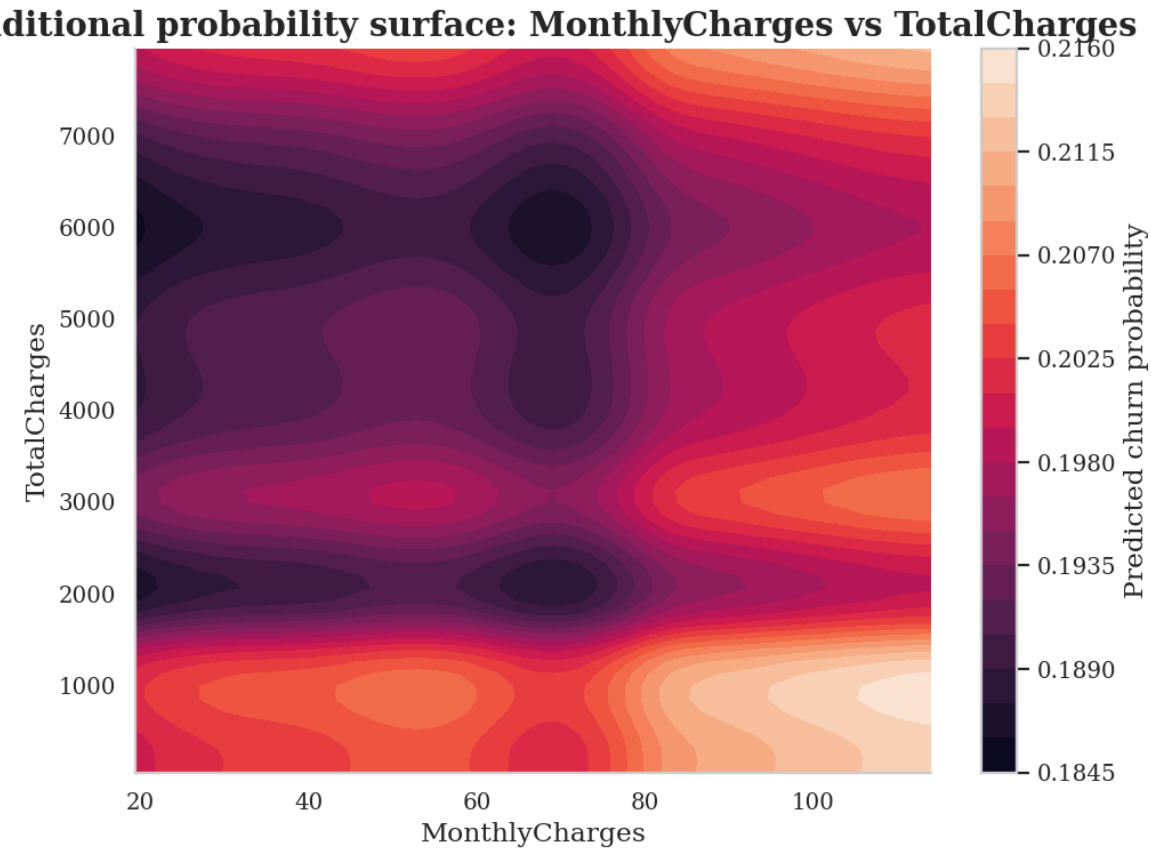
```
        raise KeyError(
            f"Both features must be in realmlp_bundle['NUMS']. "
            f"Missing: {[f for f in [feature_x, feature_y] if f not in
feat_to_idx]}"
        )

    X_num_full = realmlp_bundle["X_num"].astype(np.float32)
    X_cat_full = realmlp_bundle["X_cat"].astype(np.int64)

    jx = feat_to_idx[feature_x]
    jy = feat_to_idx[feature_y]

    x_col = X_num_full[:, jx].astype(np.float64)
    y_col = X_num_full[:, jy].astype(np.float64)

    xs = np.linspace(np.quantile(x_col, q_low), np.quantile(x_col,
q_high), grid_n)
    ys = np.linspace(np.quantile(y_col, q_low), np.quantile(y_col,
q_high), grid_n)

    # reference point = column medians in numeric input space,
    # modal category codes in categorical input space
    ref_num = np.median(X_num_full, axis=0).astype(np.float32)
    ref_cat = np.zeros(X_cat_full.shape[1], dtype=np.int64)
    for j in range(X_cat_full.shape[1]):
        vals, counts = np.unique(X_cat_full[:, j], return_counts=True)
        ref_cat[j] = vals[np.argmax(counts)]

    # build grid
    X_num_grid = np.repeat(ref_num[None, :], grid_n * grid_n, axis=0)
    X_cat_grid = np.repeat(ref_cat[None, :], grid_n * grid_n, axis=0)

    # write the 2D slice
    xx, yy = np.meshgrid(xs, ys)
    X_num_grid[:, jx] = xx.ravel().astype(np.float32)
    X_num_grid[:, jy] = yy.ravel().astype(np.float32)

    out = predict_realmlp_from_arrays(
        X_num_grid,
        X_cat_grid,
        realmlp_bundle,
        fold_cache,
        eval_bs=eval_bs,
    )
    z = out["mean_pred"].reshape(len(ys), len(xs))

    fig, ax = plt.subplots(figsize=(8, 6))
    cf = ax.contourf(xs, ys, z, levels=20)
    ax.contour(xs, ys, z, levels=[0.5], linewidths=2)
    ax.set_xlabel(feature_x)
```

```
    ax.set_ylabel(feature_y)
    ax.set_title(f"Conditional probability surface: {feature_x} vs
{feature_y}")
    plt.colorbar(cf, ax=ax, label="Predicted churn probability")
    return fig, ax


surface_features = [
    ("tenure", "MonthlyCharges"),
    ("MonthlyCharges", "TotalCharges"),
    ("tenure", "x2"),
    ("x1_0", "x2"),
]

for a, b in surface_features:
    if a in realmlp_bundle["NUMS"] and b in realmlp_bundle["NUMS"]:
        fig, ax = plot_realmlp_surface_arrays(
            realmlp_bundle=realmlp_bundle,
            fold_cache=fold_cache,
            feature_x=a,
            feature_y=b,
            grid_n=60,
            q_low=0.02,
            q_high=0.98,
            eval_bs=32768,
        )
        savefig(cfg, f"realmlp_surface_{a}_{b}.png".replace("/",
"_").replace(" ", "_"))
        plt.show()
```

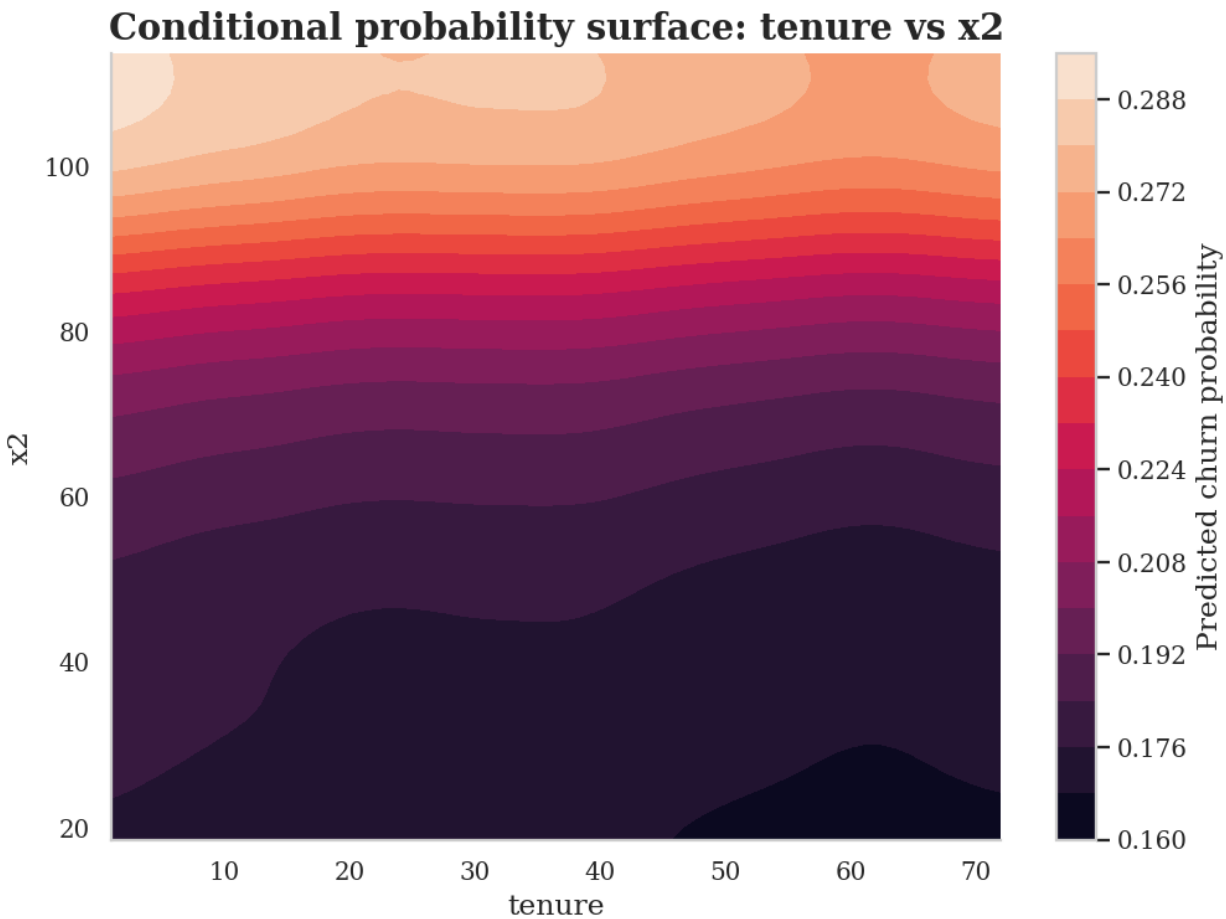
```
saved: analysis_outputs\realmlp_surface_tenure_MonthlyCharges.png
```



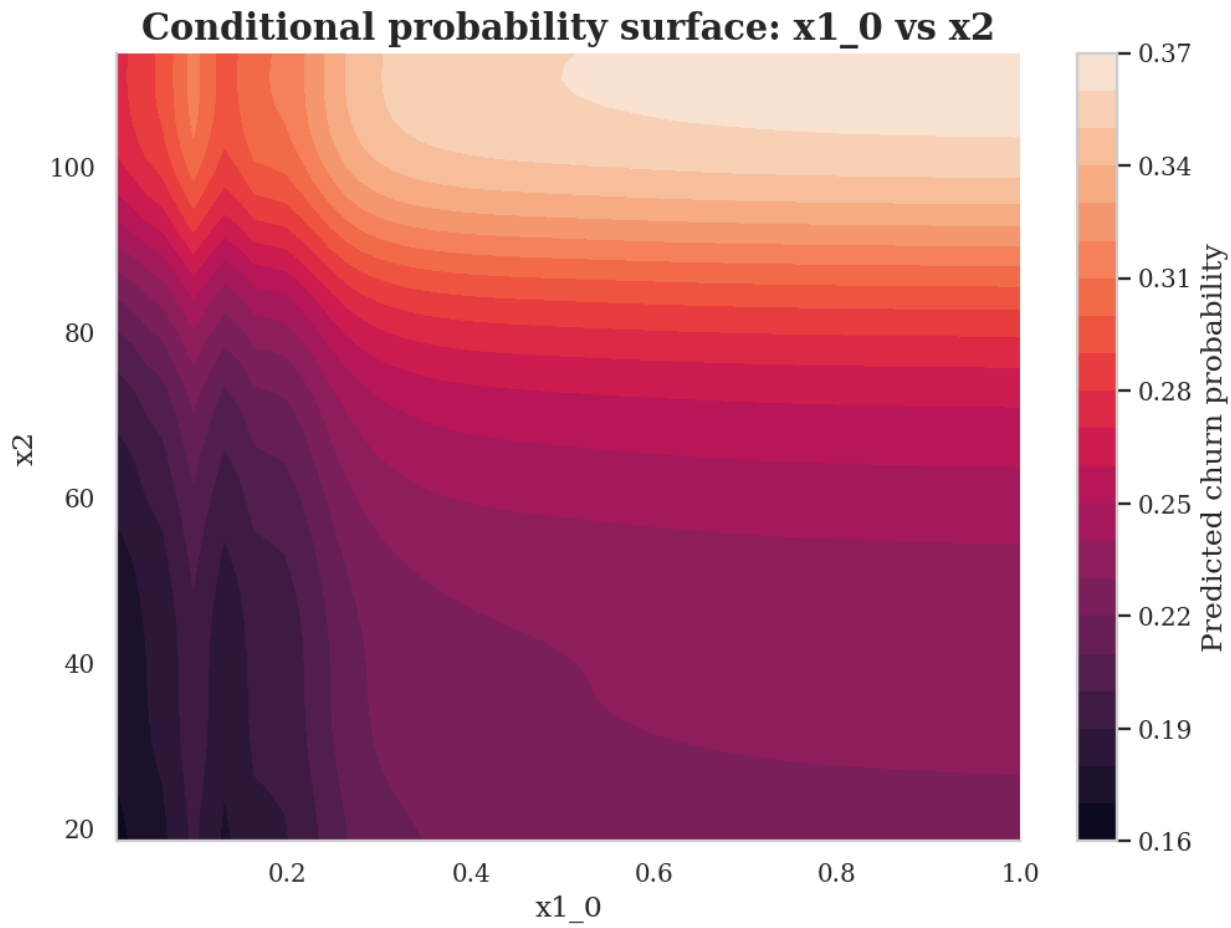

```
saved: analysis_outputs\
realmlp_surface_MonthlyCharges_TotalCharges.png
```




```
saved: analysis_outputs\realmlp_surface_tenure_x2.png
```




```
saved: analysis_outputs\realmlp_surface_x1_0_x2.png
```



Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

10. Disagreement and uncertainty structure

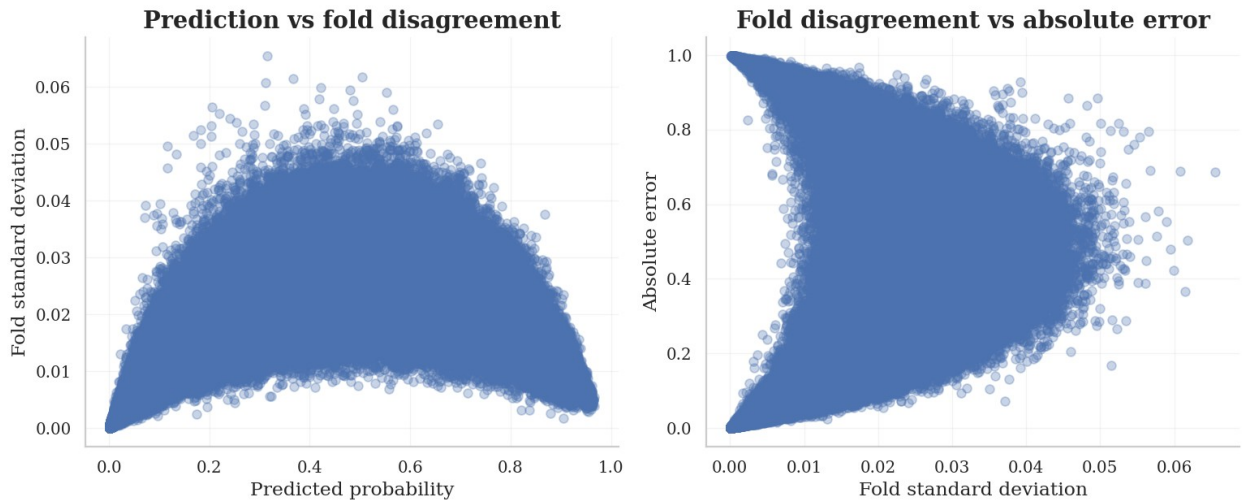
Fold-model averaging gives us a natural uncertainty proxy: disagreement across fold-members.

We inspect where disagreement is highest and whether high disagreement aligns with higher error.

```
# full train-side member prediction matrix across all fold models
train_member_out = predict_realmlp_from_arrays(
    realmlp_bundle["X_num"].astype(np.float32),
    realmlp_bundle["X_cat"].astype(np.int64),
    realmlp_bundle,
    fold_cache,
    eval_bs=32768,
)
```

```
oof_fold_pred = train_member_out["member_preds"]

analysis_train["fold_std"] = oof_fold_pred.std(axis=1)
analysis_train["fold_range"] = oof_fold_pred.max(axis=1) -
oof_fold_pred.min(axis=1)

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

axes[0].scatter(analysis_train["realmlp_pred"],
analysis_train["fold_std"], alpha=0.3)
axes[0].set_title("Prediction vs fold disagreement")
axes[0].set_xlabel("Predicted probability")
axes[0].set_ylabel("Fold standard deviation")

axes[1].scatter(analysis_train["fold_std"],
analysis_train["abs_error"], alpha=0.3)
axes[1].set_title("Fold disagreement vs absolute error")
axes[1].set_xlabel("Fold standard deviation")
axes[1].set_ylabel("Absolute error")

plt.tight_layout()
savefig(cfg, "realmlp_05_disagreement_vs_error.png")
plt.show()

uncertainty_summary = analysis_train.groupby(
    pd.qcut(analysis_train["fold_std"], q=5, duplicates="drop")
).agg(
    mean_pred=("realmlp_pred", "mean"),
    observed_rate=("y_true", "mean"),
    mean_abs_error=("abs_error", "mean"),
    n=("y_true", "size"),
)
display(uncertainty_summary)
```

```
saved: analysis_outputs\realmlp_05_disagreement_vs_error.png
```




```
                          mean_pred  observed_rate  mean_abs_error
n
fold_std

(-0.0009382, 0.000666]   0.003392       0.001750        0.005128
118839
(0.000666, 0.00317]      0.012918       0.011335        0.023767
118839
(0.00317, 0.0136]        0.147528       0.151021        0.138210
118838
(0.0136, 0.0233]         0.411039       0.442700        0.354451
118839
(0.0233, 0.0656]         0.474269       0.519232        0.449057
118839
```

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

11. Local case studies

We create representative local diagnostics for:

- extreme TP,
- extreme TN,
- high-confidence FP,
- low-score FN.

For the neural model, the most honest local explanation is a local perturbation report:

- base prediction,
- top finite-difference sensitivities for numerical variables,

- effect of grouped categorical ablations for that row.

```
def local_numeric_sensitivity_for_row_arrays(
    row_index,
    realmlp_bundle,
    fold_cache,
    features,
    n_top=12,
    eval_bs=32768,
):
    X_num_full = realmlp_bundle["X_num"].astype(np.float32)
    X_cat_full = realmlp_bundle["X_cat"].astype(np.int64)
    feature_names = list(realmlp_bundle["NUMS"])
    feat_to_idx = {f: i for i, f in enumerate(feature_names)}

    valid_features = [f for f in features if f in feat_to_idx]

    X_num_row = X_num_full[[row_index]].copy()
    X_cat_row = X_cat_full[[row_index]].copy()

    base = predict_realmlp_from_arrays(
        X_num_row,
        X_cat_row,
        realmlp_bundle,
        fold_cache,
        eval_bs=eval_bs,
    )["mean_pred"][0]

    rows = []
    for feat in valid_features:
        j = feat_to_idx[feat]

        col_all = X_num_full[:, j].astype(np.float64)
        step = max(float(np.std(col_all, ddof=0)) * 0.05, 1e-3)

        X_num_plus = X_num_row.copy()
        X_num_minus = X_num_row.copy()
        X_num_plus[0, j] = X_num_plus[0, j] + step
        X_num_minus[0, j] = X_num_minus[0, j] - step

        p_plus = predict_realmlp_from_arrays(
            X_num_plus,
            X_cat_row,
            realmlp_bundle,
            fold_cache,
            eval_bs=eval_bs,
        )["mean_pred"][0]

        p_minus = predict_realmlp_from_arrays(
            X_num_minus,
            X_cat_row,
```

```
            realmlp_bundle,
            fold_cache,
            eval_bs=eval_bs,
        )["mean_pred"][0]

        sens = (p_plus - p_minus) / (2.0 * step)

        rows.append({
            "feature": feat,
            "local_abs_sensitivity": float(abs(sens)),
            "local_signed_sensitivity": float(sens),
            "step": float(step),
            "base_value": float(X_num_row[0, j]),
        })

    out = (
        pd.DataFrame(rows)
        .sort_values("local_abs_sensitivity", ascending=False)
        .head(n_top)
        .reset_index(drop=True)
    )
    return base, out


candidate_rows = []
for q in [
    analysis_train.query("case_type ==
'TP'").sort_values("realmlp_pred", ascending=False),
    analysis_train.query("case_type ==
'TN'").sort_values("realmlp_pred", ascending=True),
    analysis_train.query("case_type ==
'FP'").sort_values("realmlp_pred", ascending=False),
    analysis_train.query("case_type ==
'FN'").sort_values("realmlp_pred", ascending=True),
]:
    if len(q):
        candidate_rows.append(int(q.index[0]))

for ridx in candidate_rows:
    base_pred, local_sens = local_numeric_sensitivity_for_row_arrays(
        row_index=ridx,
        realmlp_bundle=realmlp_bundle,
        fold_cache=fold_cache,
        features=[f for f in ["tenure", "MonthlyCharges",
"TotalCharges", "x1_0", "x1_1", "x2", "x3", "x4"]
                  if f in realmlp_bundle["NUMS"]],
        n_top=12,
        eval_bs=32768,
    )
    print({
```

```
        "row_index": ridx,
        "y_true": int(analysis_train.loc[ridx, "y_true"]),
        "pred": float(base_pred),
        "case_type": analysis_train.loc[ridx, "case_type"],
    })
    display(local_sens)
```

```
{'row_index': 213194, 'y_true': 1, 'pred': 0.9651018381118774,
'case_type': 'TP'}
```

```
          feature  local_abs_sensitivity  local_signed_sensitivity  \
0              x3           2.148020e-02             -2.148020e-02
1              x2           4.050195e-04              4.050195e-04
2            x1_0           2.104229e-04              2.104229e-04
3            x1_1           5.520270e-05             -5.520270e-05
4          tenure           5.156148e-05             -5.156148e-05
5  MonthlyCharges           2.183319e-05              2.183319e-05
6              x4           3.045480e-07             -3.045480e-07
7    TotalCharges           1.878855e-07             -1.878855e-07

         step  base_value
0    0.016756    1.000001
1    1.998573   95.449905
2    0.011330    1.000000
3    1.255739    1.000000
4    1.253095    1.000000
5    1.553371   95.449997
6   96.291824    1.000000
7  117.695736   95.449997
```

```
{'row_index': 284352, 'y_true': 0, 'pred': 0.0013824042398482561,
'case_type': 'TN'}
```

```
          feature  local_abs_sensitivity  local_signed_sensitivity  \
0              x3           7.110491e-04             -7.110491e-04
1            x1_0           1.442137e-04             -1.442137e-04
2              x2           6.622617e-06              6.622617e-06
3            x1_1           5.725560e-07              5.725560e-07
4          tenure           4.450481e-07              4.450481e-07
5  MonthlyCharges           3.327499e-07             -3.327499e-07
6              x4           5.425983e-08             -5.425983e-08
7    TotalCharges           3.971172e-08             -3.971172e-08

         step   base_value
0    0.016756     1.066182
1    0.011330     0.017770
2    1.998573    61.434166
3    1.255739    56.275570
4    1.253095    60.000000
5    1.553371    65.500000
```

```
6   96.291824  3600.000000
7  117.695736  3686.050049
```

```
{'row_index': 106705, 'y_true': 0, 'pred': 0.9608534574508667,
'case_type': 'FP'}
```

```
          feature  local_abs_sensitivity  local_signed_sensitivity  \
0              x3           3.271386e-02             -3.271386e-02
1              x2           4.718989e-04              4.718989e-04
2            x1_0           2.262046e-04              2.262046e-04
3            x1_1           6.467212e-05             -6.467212e-05
4          tenure           6.240651e-05             -6.240651e-05
5  MonthlyCharges           2.488370e-05              2.488370e-05
6              x4           3.549965e-07             -3.549965e-07
7    TotalCharges           2.271338e-07             -2.271338e-07

         step  base_value
0    0.016756    1.004212
1    1.998573   94.999908
2    0.011330    1.004210
3    1.255739    0.995807
4    1.253095    1.000000
5    1.553371   95.400002
6   96.291824    1.000000
7  117.695736   95.000000
```

```
{'row_index': 81118, 'y_true': 1, 'pred': 0.0022984042298048735,
'case_type': 'FN'}
```

```
          feature  local_abs_sensitivity  local_signed_sensitivity  \
0              x3           1.687360e-03             -1.687360e-03
1            x1_0           9.612771e-04             -9.612771e-04
2              x2           8.716414e-06              8.716414e-06
3  MonthlyCharges           3.982057e-06              3.982057e-06
4            x1_1           1.440753e-06              1.440753e-06
5          tenure           2.005759e-07              2.005759e-07
6    TotalCharges           7.149267e-08              7.149267e-08
7              x4           5.960294e-09              5.960294e-09

         step   base_value
0    0.016756     0.911324
1    0.011330     0.023982
2    1.998573    22.494736
3    1.553371    20.500000
4    1.255739    41.697559
5    1.253095    38.000000
6  117.695736   854.799988
7   96.291824  1444.000000
```

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

12. Export tables

This section writes the central report artefacts for the RealMLP study.

```
outdir = Path(cfg.output_dir)

analysis_train.to_csv(outdir / "realmlp_analysis_train.csv",
index=False)
analysis_test.to_csv(outdir / "realmlp_analysis_test.csv",
index=False)
replay["realmlp_fold_metrics_df"].to_csv(outdir /
"realmlp_fold_metrics.csv", index=False)
replay["schema_df"].to_csv(outdir / "realmlp_schema_check.csv",
index=False)
dec_realmlp.to_csv(outdir / "realmlp_deciles.csv", index=False)
perm_imp.to_csv(outdir / "realmlp_permutation_importance.csv",
index=False)
sens_df.to_csv(outdir / "realmlp_finite_difference_sensitivity.csv",
index=False)
ablation_df.to_csv(outdir / "realmlp_grouped_ablation.csv",
index=False)
uncertainty_summary.to_csv(outdir / "realmlp_uncertainty_summary.csv")

summary = {
    "oof_auc": float(replay["realmlp_oof_auc"]),
    "oof_logloss": float(log_loss(bundle["y"],
replay["realmlp_oof"])),
    "oof_brier": float(brier_score_loss(bundle["y"],
replay["realmlp_oof"])),
    "n_numeric_features": int(schema["numeric_count"]),
    "n_categorical_features": int(schema["categorical_count"]),
    "top10_permutation_importance":
perm_imp.head(10).set_index("feature")["mean_importance"].to_dict(),
    "top10_finite_difference_sensitivity":
sens_df.head(10).set_index("feature")
["mean_abs_sensitivity"].to_dict(),
}
(outdir / "realmlp_xai_summary.json").write_text(json.dumps(summary,
indent=2))
summary
```

```
{'oof_auc': 0.9171993209286087,
 'oof_logloss': 0.29869582337069117,
```

```
 'oof_brier': 0.0955834612250328,
 'n_numeric_features': 22,
 'n_categorical_features': 54,
 'top10_permutation_importance': {'x2': 0.0056478004188659225,
  'SeniorCitizen': 0.0013062768447774653,
  'x1_0': 0.00098563323021994,
  'x3': 0.0005300559637209101,
  'tenure_digit1': 0.00031854286668386855,
  'MonthlyCharges': 0.00020172727197842466,
  'TotalCharges': 0.00019811209607401015,
  'TotalCharges_digit2': 0.00018454198728912186,
  'MonthlyCharges_digit2': 6.568548744279035e-05,
  'TotalCharges_digit3': 5.456684662719091e-05},
 'top10_finite_difference_sensitivity': {'x1_0': 0.1098923310637474,
  'SeniorCitizen': 0.022135356441140175,
  'MonthlyCharges_digit2': 0.003959360998123884,
  'tenure_digit1': 0.0025660244282335043,
  'x2': 0.0012736959615722299,
  'TotalCharges_digit2': 0.0011614697286859155,
  'TotalCharges_digit3': 0.0006937597063370049,
  'TotalCharges_digit1': 0.000672676251269877,
  'MonthlyCharges_digit-1': 0.00042705165105871856,
  'TotalCharges_digit-1': 0.0003416793479118496}}
```